

Name: Y.Sreenadh

Reg no: 192525166

22. Best Fit Memory Allocation

Code:

```
#include <stdio.h>

int main() {
    int n, m, block[10], process[10];

    printf("Enter number of blocks: ");
    scanf("%d", &n);

    printf("Enter block sizes:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", &block[i]);

    printf("Enter number of processes: ");
    scanf("%d", &m);

    printf("Enter process sizes:\n");
    for(int i = 0; i < m; i++)
        scanf("%d", &process[i]);

    for(int i = 0; i < m; i++)
    {
        int pos =
-1;

        for(int j = 0; j < n; j++)
        {
            if(block[j] >= process[i] &&
                (pos == -1 || block[j] < block[pos]))
```

```

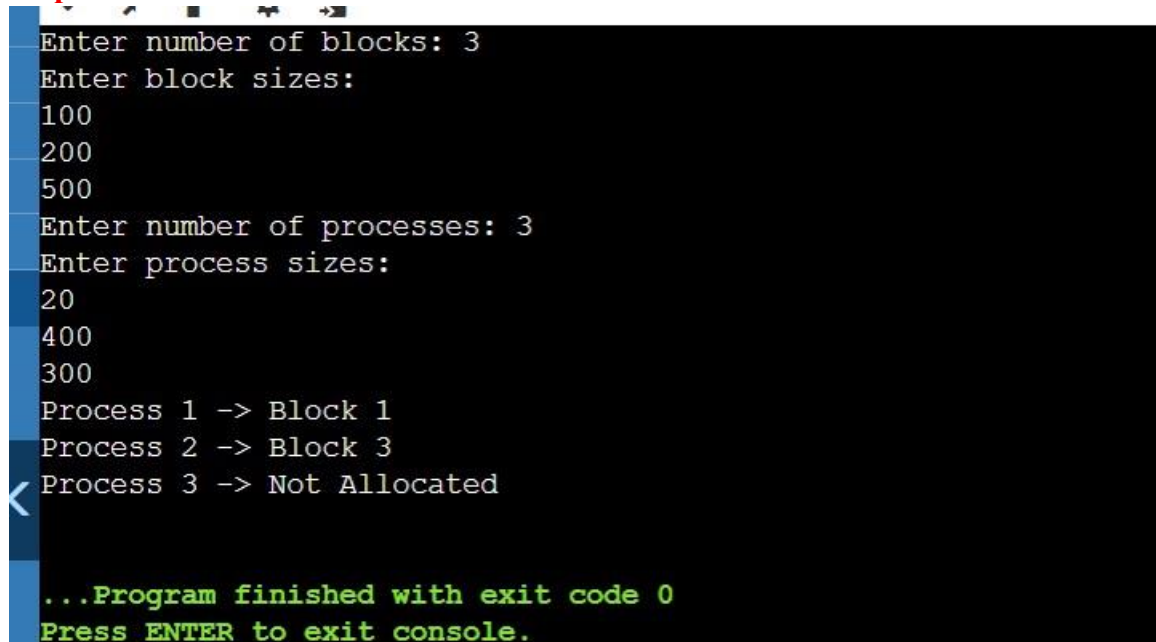
        pos = j;
    }

    if(pos != -1)
    {
        printf("Process %d -> Block %d\n", i + 1, pos + 1);
        block[pos] -= process[i];
    }
    else
        printf("Process %d -> Not Allocated\n", i + 1);
}

return 0;
}

```

Output:



```

Enter number of blocks: 3
Enter block sizes:
100
200
500
Enter number of processes: 3
Enter process sizes:
20
400
300
Process 1 -> Block 1
Process 2 -> Block 3
Process 3 -> Not Allocated

...Program finished with exit code 0
Press ENTER to exit console.

```